

Foundation (Jalapeno)

Q1. Solve the system: $x + y = 4$, $x - y = 2$. What is x ?

- (A) 1
- (B) 2
- (C) 3
- (D) 5
- (E) 6

Q2. In a basketball game, x points were scored from field goals and y points from free throws. If $x + y = 80$ and $x = 1.5y$, what is y ?

- (A) 20
- (B) 25
- (C) 30
- (D) 32
- (E) 40

Q3. A tennis player wins $2x$ points from aces and $3y$ points from volleys. If $2x + 3y = 50$ and $y = 5$, what is x ?

- (A) 5
- (B) 7.5
- (C) 10
- (D) 12.5
- (E) 15

Q4. Solve the system: $y = 3x - 1$, $y = 2x + 1$. What is y ?

- (A) 2
- (B) 4
- (C) 5
- (D) 6
- (E) 8

Q5. A cyclist travels x miles uphill and y miles downhill. If $x + y = 20$ and $x = y + 4$, what is y ?

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 10

Q6. Solve the system: $2x + 2y = 8$, $x - y = -2$. What is y ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q7. A football team scores x touchdowns and y field goals. If $x + y = 6$ and $x = 2y$, what is y ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 6

Q8. Solve the system: $x + y = 7$, $2x - 2y = -2$. What is y ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Open Ended Questions (FRQ)

Q9. Solve the system: $x + 2y = 10$, $3x - 2y = 4$. What are x and y ?

Q10. A swimmer swims x laps in the morning and y laps in the evening. If $x + y = 15$ and $2x - y = 3$, what are x and y ?

Chef's Tips

- Tip 1: Ensure students have a firm grasp of substitution and elimination methods.
- Tip 2: Encourage students to check their solutions by plugging them back into the original equations.
- Tip 3: Remind students that the solution to a system can be one point, no points, or infinitely many points.